

EX.NO:03 Exploring Different Types of Prompts

Aim:

To design prompts using Straightforward, Tabular Format, Preceding Question, and Missing Word prompting techniques and generate results for each.

Theory/Explanation:

Prompt engineering helps in generating accurate and structured responses from AI by designing inputs properly. Different prompt types are used for different purposes such as direct answers, structured outputs, conversational flow, and knowledge testing.

Procedure:

1. Select different use cases (Cybersecurity, AI, Networking, etc.)
 2. Create two prompts for each prompt type
 3. Generate outputs for each prompt
 4. Analyze clarity and correctness
 5. Compile results
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1. Straightforward Prompts

(Simple and direct instructions)

Prompt 1:

"Explain what a firewall is in one sentence."

Result:

A firewall is a security system that monitors and controls incoming and outgoing network traffic.

Prompt 2:

"List three advantages of blockchain technology."

Result:

Decentralization, transparency, and high security.

2. Tabular Format Prompting

(Structured output in table form)

Prompt 1:

"Compare Java, Python, and C++ based on speed, usage, and difficulty."

Result:

Language	Speed	Usage	Difficulty
Java	Medium	Enterprise apps	Medium
Python	Slow	AI, Data Science	Easy
C++	Fast	System Programming	Hard

Prompt 2:

"Create a table showing LAN, MAN, and WAN with features and examples."

Result:

Network	Feature	Example
LAN	Small area	Home network
MAN	City coverage	Cable network
WAN	Large area	Internet

3. Preceding Question Prompting

(Starts with a question, then explanation)

Prompt 1:

"What is Artificial Intelligence? Explain how it is used in daily life."

Result:

Artificial Intelligence is the ability of machines to perform tasks that require human intelligence. It is used in daily life through virtual assistants, recommendation systems, and automation tools.

Prompt 2:

"Why is phishing dangerous? Describe how phishing attacks work."

Result:

Phishing is dangerous because it steals sensitive information like passwords and bank details. Attackers send fake emails or websites that look real to trick users into entering their data.

4. Missing Word Prompting

(Fill-in-the-blank prompts)

Prompt 1:

"The brain of the computer is called the ____."

Result:

CPU

Prompt 2:

"HTTP stands for HyperText Transfer ____."

Result:

Protocol

Result:

All four types of prompts—Straightforward, Tabular, Preceding Question, and Missing Word—were successfully created and tested. The outputs were clear, relevant, and suitable for different use cases.

Conclusion:

Different prompting techniques improve the quality of AI responses by making them more structured, clear, and context-aware. These techniques are useful in real-world applications like chatbots, education, and automation systems.